

Lucas Lange

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Education

- 10/2023 - Present **Technical University of Munich (Munich, Germany)**
M.Sc. Computer Science
Major in Machine Learning
Minors in High Performance Computing and Databases
Erasmus Semester at University of Agder (UiA), Norway
- 10/2018 - 09/2023 **Technical University of Munich (Munich, Germany)**
B.Sc. Computer Science with a minor in Psychology

Experience

- 05/2026 - Present **Agentic AI Engineering (Master's Thesis) at BMW Group**
- Architected 3 agent systems from scratch - monolithic (Google ADK), hierarchical multi-agent (coordinator + specialist sub-agents), and state-graph workflow (LangGraph) - connected via A2A protocol and MCP, enabling the first empirical architecture comparison for tool-calling enterprise agents
 - Designed a 7-phase experimental pipeline with 500+ automated evaluation runs across 20 models, producing statistically validated results with confidence intervals and Holm-Bonferroni significance testing, plus automated cost/performance Pareto analysis via LiteLLM routing
 - Built a dual LLM-as-judge scoring system (GPT-4o + Claude) combined with deterministic tool-call evaluators (precision, recall, order, reliability), running in an async Python pipeline with scenario-driven test fixtures for fully reproducible benchmarking
- 10/2025 - 01/2026 **Meta Team Lead / DevOps Engineer at Adtran**
- Owned DevOps lifecycle (CI/CD, Docker, hosting) and implemented offline-first sync with conflict resolution for a fault-management platform (Rust, PostgreSQL, React Native)
- 09/2024 - 12/2024 **Software Engineer Intern at BMW Group**
- Developed a Python-based anomaly detection system with REST APIs, collaborating with UI designers on FIGMA prototypes for the frontend interface, achieving a 10% error rate reduction in production configurations
 - Engineered key components of an intelligent search algorithm using Java Quarkus, Python, and Kafka Streams, while implementing CI/CD pipelines in a joint BMW-Microsoft team, contributing to a solution achieving €9M annual cost savings
- 03/2024 - 08/2024 **Machine Learning Working Student at OrthoCoPilot**
- Designed and implemented a RAG pipeline (LangChain, Llama 2) to extract and structure medical information from clinical documentation, collaborating with medical professionals to ensure clinical accuracy
- 09/2023 - 12/2023 **Machine Learning Working Student at UniCredit**
- Built data wrangling and preprocessing pipelines for a sentiment analysis dataset, aligning deliverables directly with stakeholders (Python, data pipeline, NLP)
- 11/2022 - 06/2023 **Data Analytics and Machine Learning Working Student at Nebumind**
- Automated an image-based anomaly detection pipeline and built the accompanying dashboard UI (Python, computer vision, dashboard)

Skills

- IT Agentic AI, LLM, RAG, Machine Learning, DL, LangChain, LangGraph, LiteLLM, OpenAI API
Python, PyTorch, Java (Quarkus), SQL, Kafka, Azure
- Languages Linux, Git, GitHub, Docker, Atlassian Suite, CI/CD
- Interests English (Native), German (Native), French (Bilingual)
Rowing, Hiking